MCC Entrance Test, August 2026 — Day 1 — Solutions

Level T1

Test Information. Level T1 — 90 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra (×2), Combinatorics (×2), Geometry, Number Theory.
Rules: show your work (partial credit rewards genuine progress); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts).[1] A traveler crosses three toll bridges, one after another. Crossing any bridge doubles the number of coins in the traveler's pouch; immediately afterward, the traveler pays a toll of 24 coins. After paying the toll at the third bridge, the traveler's pouch is completely empty. How many coins did the traveler have before crossing the first bridge?

Official solution. Work backward from the end. After the 3rd toll, the traveler has 0 coins, so right before that toll they had $0 + 24 = 24$. That 24 was itself the result of the 3rd doubling, so right after the 2nd toll they had $24 \div 2 = 12$.

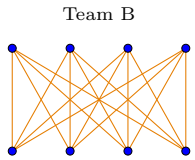
Before the 2nd toll: $12 + 24 = 36$; before the 2nd doubling (right after the 1st toll): $36 \div 2 = 18$.

Before the 1st toll: $18 + 24 = 42$; before the 1st doubling – the traveler's starting amount – $42 \div 2 = \boxed{21}$.

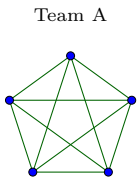
Check forward: $21 \xrightarrow{\times 2} 42 \xrightarrow{-24} 18 \xrightarrow{\times 2} 36 \xrightarrow{-24} 12 \xrightarrow{\times 2} 24 \xrightarrow{-24} 0$. ✓ □

Problem 2 (10 pts).[2] Two teams, Team A and Team B, each have exactly 20 members and take part in a competition. When the competition ends, every member of Team A shakes hands with every member of Team B, once each. Afterward, every member of Team A also shakes hands with each of their own teammates, once each. If there are n handshakes between the two teams and m handshakes within Team A, what is the value of $n + m$?

Official solution. **Between-team handshakes.** Every one of the 20 members of Team A meets every one of the 20 members of Team B – two *different* groups, so no handshake gets counted twice: $n = 20 \times 20 = 400$.



every Team A–Team B pair shakes hands *once*: no double count



each edge is one within-team handshake, counted once from *either* end – so $\binom{n}{2} = \frac{n(n-1)}{2}$, not $n(n-1)$

(illustrated with 4 members per team; the actual problem has 20)

Within-team handshakes. The 20 members of Team A shake hands only *within their own team*. Multiplying 20×19 would count "A shakes B's hand" and "B shakes A's hand" as two separate events, but they are the same handshake, so the count must be divided by 2:

$$m = \frac{20 \times 19}{2} = 190 \implies n + m = 400 + 190 = \boxed{590}.$$

□

Problem 3 (15 pts).[3] A fund starts at exactly 1000 dollars. Each year, on the same date, the fund's value changes by a fixed percentage: Year 1: the value increases by 20%. Year 2: the value decreases by 50%. Year 3: the value increases by 30%. Year 4: the value increases by 40%. What is the fund's value:

(a) right after the Year 2 change takes effect (that is, after the first two changes)? [6 pts]
(b) right after the Year 4 change takes effect (that is, after all four changes)? [9 pts]

[1] Kangaroo Math Competition (Kangourou Sans Frontières). [2] MATHCOUNTS. [3] Purple Comet! Math Meet, Middle School Division.

Official solution. **Part (a).** After Year 1: $1000 \times 1.20 = 1200$. After Year 2: $1200 \times 0.50 = \boxed{600}$.

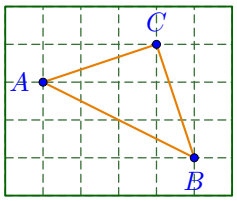
Part (b). After Year 3: $600 \times 1.30 = 780$. After Year 4: $780 \times 1.40 = \boxed{1092}$.

(Check directly: $1000 \times 1.2 \times 0.5 \times 1.3 \times 1.4 = 1000 \times 1.092 = 1092$.) □

Problem 4 (15 pts).[4] On a coordinate grid, a rectangle has corners at $(0,0)$, $(6,0)$, $(6,5)$, and $(0,5)$. A triangle is drawn inside it with vertices $A = (1,3)$, $B = (5,1)$, $C = (4,4)$.

(a) Find the area of the smallest rectangle, with sides parallel to the grid lines, that completely contains triangle ABC. [4 pts]

(b) Find the area of triangle ABC, and hence find what fraction of the large rectangle's area is covered by the triangle. [11 pts]



Official solution. **Part (a).** The smallest enclosing rectangle spans x from 1 to 5 (width 4) and y from 1 to 4 (height 3): corners $(1,1)$, $(5,1)$, $(5,4)$, $(1,4)$. Notice $B = (5,1)$ is itself a corner of this box, while A and C sit on two of its edges. Area $= 4 \times 3 = \boxed{12}$.

Part (b). Subtract the three right-triangle corners of the box that lie outside triangle ABC:

- Top-left corner $(1,4)$–A–C: legs 1 and 3, so area is 1.5.
- Bottom-left corner $(1,1)$–A–B: legs 2 and 4, so area is 4.
- Top-right corner $(5,4)$–C–B: legs 1 and 3, so area is 1.5.

Total removed $= 1.5 + 4 + 1.5 = 7$, so the triangle's area is $12 - 7 = \boxed{5}$. The whole grid rectangle has area $6 \times 5 = 30$, so the triangle covers $\frac{5}{30} = \boxed{\frac{1}{6}}$ of it. □

Problem 5 (25 pts).[5] A collection of 11 tokens consists of 8 silver tokens, each worth 10 points, and 3 gold tokens, each worth 25 points. A non-empty subset of these 11 tokens is chosen, and its value is the sum of the point-values of the tokens in it.

(a) Suppose the subset uses *only* silver tokens (no gold ones). List every possible value, using at least one silver token. How many different values are possible? [6 pts]

(b) Suppose the subset uses exactly one gold token together with any number of silver tokens (0 to 8 of them). Explain why every value made this way ends in the digit 5, while every value made using only silver tokens ends in the digit 0. [4 pts]

(c) Taking into account ALL non-empty subsets of the 11 tokens (any combination of silver and gold), how many different total values are possible in all? [15 pts]

Official solution. **Part (a).** With d silver tokens ($d = 1, \ldots, 8$), the value is $10d$: $10, 20, 30, 40, 50, 60, 70, 80$ – $\boxed{8}$ different values.

Part (b). A subset with q gold tokens and d silver tokens has value $25q + 10d$. If q is odd (1 or 3), $25q$ ends in 5 (25 or 75), and adding any multiple of 10 never changes the last digit – so the value ends in 5. If q is even (0 or 2), $25q$ ends in 0 (0 or 50), so the value ends in 0. Every value falls into exactly one of these two families, and the families never overlap.

Part (c). Since the two families from part (b) never overlap, count each family separately and add.

Ends in 0 ($q = 0$ or $q = 2$): $q = 0$ gives $10, \ldots, 80$ (8 values); $q = 2$ gives $50, 60, \ldots, 130$ (9 values). These overlap on 50–80, so together they form one unbroken run $10, 20, \ldots, 130$: $\frac{130-10}{10} + 1 = 13$ values.

Ends in 5 ($q = 1$ or $q = 3$): $q = 1$ gives $25, 35, \ldots, 105$ (9 values); $q = 3$ gives $75, 85, \ldots, 155$ (9 values). These overlap on 75–105, so together they form $25, 35, \ldots, 155$: $\frac{155-25}{10} + 1 = 14$ values.

Total distinct values $= 13 + 14 = \boxed{27}$. □

[4] American Mathematics Competition 8 (AMC 8). [5] Gauss Contest, Centre for Education in Mathematics and Computing.

Problem 6 (25 pts).[6] When you write $\frac{5}{13}$ as a decimal, you get $0.384615384615384615\ldots$, where the block of six digits "384615" repeats forever.

(a) Using long division, verify that $5 \div 13$ produces the repeating block 384615, and explain why the decimal is guaranteed to eventually start repeating. [10 pts]

(b) Using the repeating block, predict what the 15th digit after the decimal point should be — without writing out 15 digits from scratch. Then write the decimal out far enough to check that your prediction is correct. [5 pts]

(c) What is the 2019th digit after the decimal point in the decimal expansion of $\frac{5}{13}$? [10 pts]

Official solution. **Part (a). Step 1 – do the long division.** Dividing $5.000000\ldots$ by 13, one step at a time:

$$50 \div 13 = 3 \text{ r } 11, \quad 110 \div 13 = 8 \text{ r } 6, \quad 60 \div 13 = 4 \text{ r } 8,$$
$$80 \div 13 = 6 \text{ r } 2, \quad 20 \div 13 = 1 \text{ r } 7, \quad 70 \div 13 = 5 \text{ r } 5,$$

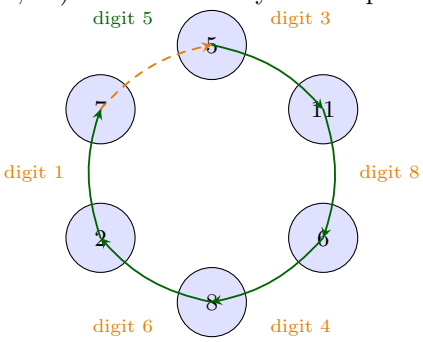
then $50 \div 13$ again – the *same* dividend (and so the same remainder, 11) as after the very first step.

Step 2 – see why it *has* to repeat. In long division, the next digit and next remainder are completely determined by the current remainder – there's no other choice involved.

So the moment a remainder repeats, every digit after it must repeat too, in exactly the same order.

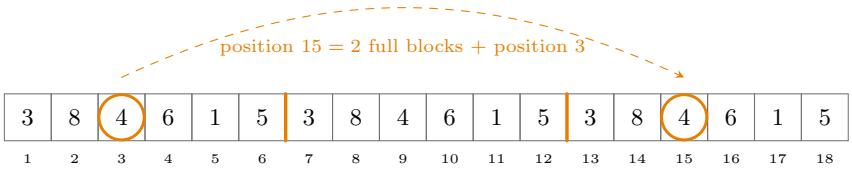
Since dividing by 13 can only ever leave a remainder from 0 through 12 (only 13 possibilities), some remainder is guaranteed to repeat within at most 13 steps – here it happens after 6.

Each arrow above is one long-division step, labeled by the digit it produces; following the arrows all the way around spells out $3, 8, 4, 6, 1, 5$ exactly, and then lands back where it started at remainder 5 – so the block $\boxed{384615}$ is guaranteed to repeat forever.



remainder comes back to 5 – so from here the digits $3, 8, 4, 6, 1, 5$ just repeat forever

Part (b). Since the block has length 6, digit 15 should match digit $(15 \bmod 6) = 3$ of the block 384615 – namely $\boxed{4}$.



Check directly: writing the decimal out to 15 places, $0.384615384615384\ldots$, the 15th digit is indeed 4. ✓

Part (c). Using the same idea as part (b): since the block has length 6, digit #2019 occupies the same position in the block as digit #$(2019 \bmod 6)$. Dividing, $2019 = 336 \times 6 + 3$, so the remainder is 3 – meaning digit #2019 matches digit #3 of the block "384615", which is $\boxed{4}$. □

[6] Math Olympiads for Elementary and Middle Schools (MOEMS).